

SAI YASWANTH YENGISETTY

AI Automation Engineer | AI Agents | LLM Applications | AWS

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PROFESSIONAL SUMMARY:

AI/ML Engineer with 2 years of experience designing, building, and deploying AI-powered applications, intelligent automation workflows, NLP systems, and scalable backend infrastructure using Python, PostgreSQL, AWS, and modern AI tooling. Skilled in building production-grade AI chatbots, Text-to-SQL systems, AI agents, RAG pipelines, and automation-focused AI applications that solve real operational and business problems.

Strong expertise in prompt engineering, embeddings, vector retrieval concepts, API integrations, cloud deployment, and backend automation workflows. Experienced in building end-to-end AI systems from rapid prototyping to scalable deployment with measurable business impact.

Passionate about AI-first engineering workflows, operational automation, scalable AI infrastructure, and building practical AI products that improve efficiency and user experience.

TECHNICAL SKILLS:

AI / LLM Engineering:	AI Agents, LLM Applications, Prompt Engineering, RAG Pipelines, Embeddings, NLP Systems, Text-to-SQL Systems, AI Automation Workflows, Retrieval Workflows, Hallucination Reduction, AI Model Evaluation, AI Workflow Optimization, Semantic Search Concepts.
Programming & Backend:	Python, SQL, FastAPI, Flask, REST APIs, Backend Automation, API Integrations.
Databases & Infrastructure:	PostgreSQL, Aurora PostgreSQL, MySQL, Redis, Query Optimization, Database Performance Tuning, Schema Design.
Machine Learning & Data:	Scikit-Learn, TensorFlow, PyTorch, Pandas, NumPy, Feature Engineering, Hyperparameter Tuning, Cross Validation, Model Evaluation, Regression, Classification
Generative AI & NLP:	LLMs, RAG Systems, Embeddings, Prompt Engineering
Cloud & Deployment:	AWS (SageMaker, EC2, Lambda, S3, RDS, Aurora, CloudWatch)
Developer Tools:	Git, GitHub, VS Code.
Frontend & UI:	JavaScript, React (Basic), Streamlit.

PROFESSIONAL EXPERIENCE:

Billion Bright Solutions — AI/ML Engineer | Remote | September 2024 – Present

Project 1: AI-Powered Chatbot for Customer Support

- Built and deployed AI-powered customer support workflows using Python, NLP techniques, embeddings, and scalable backend systems.
- Developed intelligent AI inference pipelines and engineered NLP features including TF-IDF and vector embeddings for response generation workflows.
- Designed scalable PostgreSQL schemas for storing high-volume AI conversations, workflow analytics, and operational metrics.
- Improved AI response quality through prompt optimization, model evaluation, hyperparameter tuning, and hallucination reduction techniques.
- Collaborated with operational workflows to solve business bottlenecks using AI-driven automation systems.
- Developed scalable backend APIs and automation pipelines supporting production AI workflows.

Business Impact:

- Reduced model response latency by 38%.
- Achieved 99.99% system uptime using Multi-AZ architecture.
- Reduced manual customer support workload by 40%.
- Enabled AI chatbot workflows supporting thousands of daily users.

Project 2: AI Professional Notification Generator for Rupe Ecom Apps

- Designed real-time data pipelines using Python and SQL to support ML-driven notification workflows.
- Engineered behavioral features from user engagement data to improve prediction quality.
- Built and tuned Random Forest and Logistic Regression models to predict user engagement likelihood using 5-fold cross-validation.
- Applied regularization techniques to prevent overfitting and improve generalization.
- Deployed scalable ML services backed by AWS Aurora PostgreSQL with optimized query performance.

Business Impact:

- Supported 100K+ daily active users.

- Delivered sub-second query responses enabling real-time AI notification generation.
- Improved database throughput by 42% via optimized indexing and caching strategies.
- Enabled seamless scaling of notification system during festival and sale seasons with zero downtime.

Bytehubble.AI — PostgreSQL DBA Intern

Hyderabad, India | February 2024 – July 2024

Key Responsibilities

- Managed AWS Aurora PostgreSQL clusters and automated backups to ensure high availability and data durability.
- Supported schema deployments across development, testing, and production environments.
- Implemented Role-Based Access Control (RBAC) and enforced database security best practices.
- Generated performance and query optimization reports using pgBadger to identify bottlenecks.
- Developed Python utilities for automated report generation and log analysis to support database monitoring.
- Supported analytics and ML workloads by optimizing query performance for model training datasets.

PROJECTS:

AI Job Alert Agent:

GitHub: <https://github.com/Yashvanth-Tech-AI/Job-Alert-Agent>

- Built an AI-powered job discovery and alert automation platform designed to streamline job search workflows and improve candidate-job matching efficiency.
- Developed intelligent search and recommendation workflows using NLP, embeddings, prompt engineering, and AI-assisted filtering techniques.
- Designed backend automation pipelines for collecting, processing, and categorizing job postings from multiple sources.
- Built scalable APIs and database workflows using Python and PostgreSQL to support real-time job alert generation and personalized recommendations.
- Implemented AI-driven retrieval and matching workflows to improve relevance and reduce manual job discovery effort.
- Focused on building production-style AI tooling with automation-first workflows and scalable backend architecture.
- Developed end-to-end application workflows including backend services, API integrations, and deployment-ready infrastructure.

MegatuneDB — Natural Language to SQL Engine:

- Built a production-grade Text-to-SQL system using a fine-tuned GPT-2 model trained on 9,000+ SQL query pairs.
- Developed AI workflows that convert natural language questions into executable SQL queries.
- Applied prompt engineering and schema-aware feature engineering techniques to improve SQL generation accuracy.
- Designed scalable REST-based inference APIs with Streamlit interface integration.
- Improved model reliability using cross-validation, dropout regularization, and evaluation workflows.
- Achieved 92% SQL accuracy with sub-500ms inference latency.
- Built practical AI tooling focused on improving database accessibility and operational efficiency.

AI Customer Support Automation System

- Developed an AI-powered support assistant capable of handling customer queries using NLP and retrieval-based AI workflows.
- Implemented embeddings-based response systems and backend automation pipelines.
- Designed scalable PostgreSQL logging and analytics systems for operational AI monitoring.
- Built cloud-native deployment workflows using AWS serverless services.
- Focused on reducing manual support operations and improving customer response efficiency.

Used Car Price Prediction Model (Regression ML Project):

- Developed a supervised regression model trained on 1,700+ vehicle records to predict used car prices using structured datasets.
- Performed exploratory data analysis (EDA) including correlation heatmaps, distribution analysis, and pair plots.
- Engineered features such as car age and brand ranking to enhance predictive power.
- Applied Label Encoding for categorical variables and StandardScaler for feature normalization.
- Built and compared Linear Regression and Random Forest Regressor models.
- Performed 70/30 train-test split with fixed random_state for reproducibility.
- Evaluated performance using Mean Squared Error (MSE) and R² score.
- Improved model performance using GridSearchCV with 5-fold cross-validation.
- Achieved optimized Random Forest R² score of 0.68.
- Analyzed feature importance to interpret key price-driving factors.

EDUCATION:

Bachelor of Technology (B.Tech) in Computer Science and Technology
Annamacharya Institute of Technology & Sciences, Tirupati, India
2021 – 2024 | CGPA: 7.77

CERTIFICATIONS:

- Generative AI Specialization — **Edureka**
- AWS Cloud Practitioner Essentials — **Coursera**
- Python for Data Science and Machine Learning — **Edureka**

LANGUAGES:

English | Telugu | Hindi.